Problem Analysis

The Smart Parking Space Rental & License Validator system manages parking bay reservations, vehicle number-plate validation, parking fee calculation, and unauthorized overstay detection. The main actors are the Vehicle Owner and Parking Warden. The system aims to provide accurate, efficient, and reliable parking management.

Smart Parking Space Rental & License Validator
Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record vehicle entry time via license plate scan and compute the total billing amount upon exit based on tiered hourly duration.	**High**	**Pass:** The system records the entry time from the license plate scan and calculates the correct bill according to the applicable hourly tier at exit. **Fail:** Entry time is not recorded or an incorrect billing tier is applied.	Required for accurate parking-duration tracking and billing.
FR-002	Functional	The system shall allow the Vehicle Owner to view available parking bays and reserve a bay for a specified time period.	**High**	**Pass:** A reservation is created only when the selected bay is available for the requested period. **Fail:** The system permits an overlapping reservation for an unavailable bay.	Supports real-time parking reservation and prevents double booking.

FR-003	Functional	The system shall validate the vehicle number plate captured by the gate camera against the vehicle's parking reservation before allowing entry.	**High**	**Pass:** A vehicle with a valid active reservation is accepted. **Fail:** A vehicle without a valid reservation is incorrectly granted entry.	Ensures that only authorized vehicles enter the parking facility.
FR-004	Functional	The system shall detect and flag a vehicle when it remains parked beyond its authorized reservation period.	**High**	**Pass:** The system flags the vehicle after the authorized parking period has expired. **Fail:** An overstay is not detected or flagged.	Enables the Parking Warden to identify unauthorized overstays.
FR-005	Functional	The system shall update the availability status of parking bays when a reservation, vehicle entry, or vehicle exit occurs.	**Medium**	**Pass:** The displayed availability reflects the current status of bays after each reservation, entry, or exit. **Fail:** A bay is shown as available when it is reserved or occupied.	Keeps parking availability accurate for reservations and management.
NFR-001	Nonfunctional — Performance & Security	The gate camera license plate lookup API shall verify parking reservation validity in under **300 ms**.	**High**	**Pass:** Benchmarking confirms reservation validation completes in less than 300 ms under simulated peak load and meets the required security checks. **Fail:** Validation takes 300 ms or more or fails the security checks.	Provides fast and secure vehicle validation at the parking gate.

NFR-002	Nonfunctional — Reliability	The system shall reliably process parking reservations and vehicle-validation transactions without losing or duplicating transaction records during peak usage.	**High**	**Pass:** Testing under simulated peak usage shows no lost or duplicate reservation or validation records. **Fail:** Any reservation or validation transaction is lost or duplicated.	Ensures dependable operation of the parking system during heavy usage.

Use-Case Flow

Use Case: Reserve Parking Bay

Main Success Scenario

1. Vehicle Owner selects the **"Reserve Parking Bay"** option.
2. System displays the available parking bays.
3. Vehicle Owner selects a parking bay and specifies the required parking duration.
4. System checks the availability of the selected parking bay.
5. System calculates the applicable parking rate based on the requested duration.
6. Vehicle Owner confirms the reservation.
7. System records the reservation details.
8. System marks the selected parking bay as reserved.
9. System displays a reservation confirmation to the Vehicle Owner.
10. Use case ends successfully.

Alternate Flow

4a. Selected Parking Bay Unavailable

- **4a1.** System displays a message that the selected parking bay is unavailable.
- **4a2.** System displays other available parking bays.
- **4a3.** Vehicle Owner selects another available parking bay.
- **4a4.** The flow continues from Step 4.